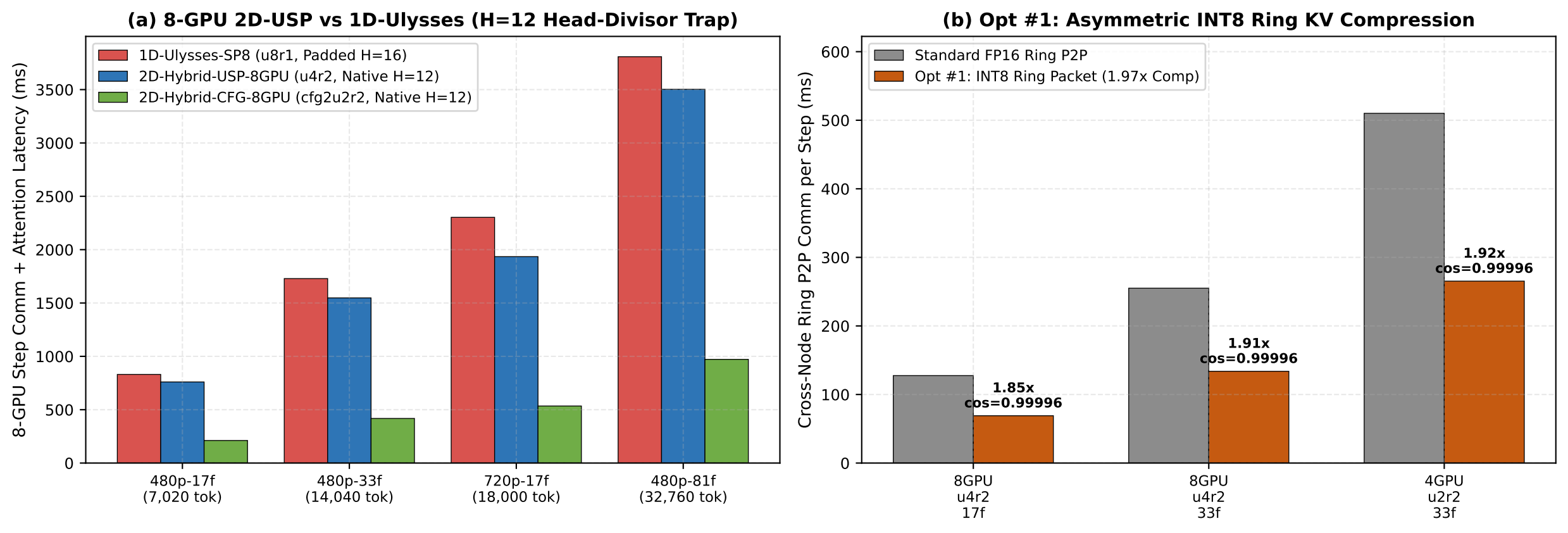
(a) 8-GPU 2D-USP vs 1D-Ulysses (H=12 Head-Divisor Trap)
8-GPU Step Comm + Attention Latency (ms)
1D-Ulysses-SP8 (u8r1, Padded H=16)
2D-Hybrid-USP-8GPU (u4r2, Native H=12)
2D-Hybrid-CFG-8GPU (cfg2u2r2, Native H=12)
0
500
1000
1500
2000
2500
3000
3500
480p-17f
(7,020 tok)
480p-33f
(14,040 tok)
720p-17f
(18,000 tok)
480p-81f
(32,760 tok)
(b) Opt #1: Asymmetric INT8 Ring KV Compression
Cross-Node Ring P2P Comm per Step (ms)
Standard FP16 Ring P2P
Opt #1: INT8 Ring Packet (1.97x Comp)
0
100
200
300
400
500
600
1.85x
cos=0.99996
1.91x
cos=0.99996
1.92x
cos=0.99996
8GPU
u4r2
17f
8GPU
u4r2
33f
4GPU
u2r2
33f